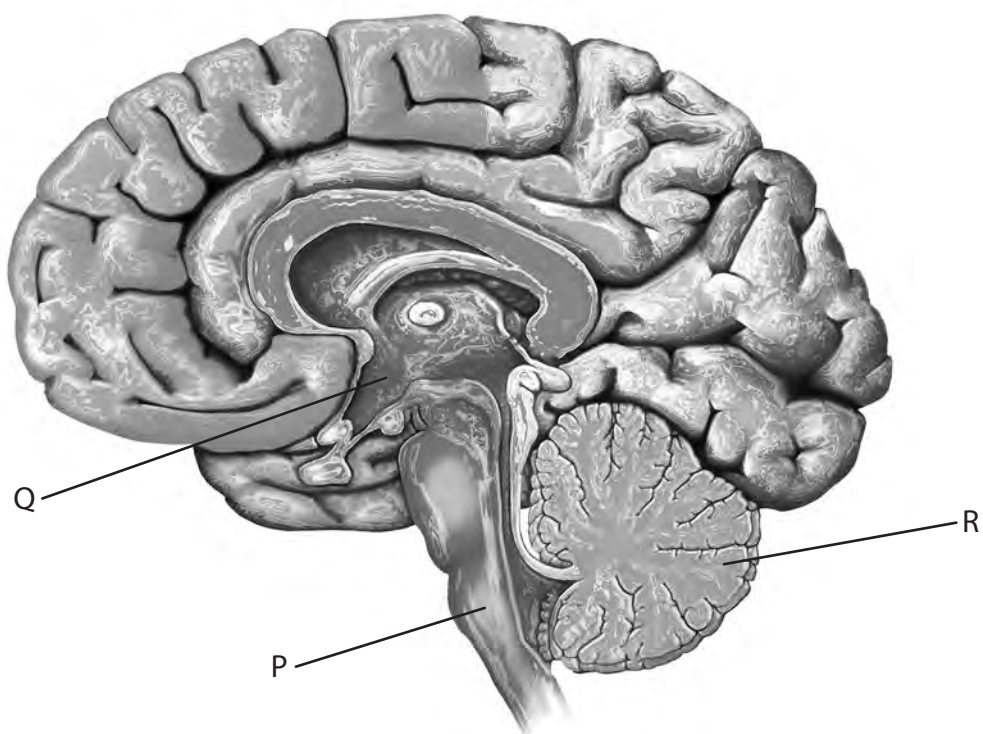


3 The brain is responsible for processing information and coordinating responses.

(a) The diagram shows a human brain.

Three regions of the brain have been labelled.



(Source: © MARK GARLICK/SCIENCE PHOTO LIBRARY)

Complete the table with the name and function of each region.

(2)

Label	Name of the region	Function of the region
P	medulla oblongata	
Q		
R		balance and coordinating movement

(b) Seizures in the brain can be investigated using different imaging techniques.

(i) Which one of the following statements is correct?

(1)

- ☐ **A** CT scans are produced by the absorbance of magnetic fields
- ☐ **B** CT scans are produced by the absorbance of radio waves by deoxyhaemoglobin
- ☐ **C** CT scans produce a cross-sectional image of a thin slice through the body
- ☐ **D** CT scans can be used to study brain activity while a person is undertaking tasks

(ii) During a seizure, part of the brain will be overactive.

Describe how positron emission tomography (PET) could be used to identify the part of the brain involved.

(3)

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(c) Some seizures can result in an increase in body temperature.

Suggest how a seizure in part of the brain could lead to an increase in body temperature.

(3)

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(Total for Question 3 = 9 marks)
